

1. The term independent of x in the expansion of $(1+x)^n \left(1+\frac{1}{x}\right)^n$ is
(a) $C_0^2 + 2C_1^2 + \dots + (n+1)C_n^2$ (b) $(C_0 + C_1 + \dots + C_n)^2$
(c) $C_0^2 + C_1^2 + \dots + C_n^2$ (d) None of these
2. The coefficient of x^n in the expansion of $(1+x)(1-x)^n$ is
(a) $(-1)^{n-1}n$ (b) $(-1)^n(1-n)$
(c) $(-1)^{n-1}(n-1)^2$ (d) $(n-1)$
3. The number of terms in the expansion of $[(x+3y)^2(3x-y)^2]^3$ is
(a) 14 (b) 28 (c) 32 (d) 56
4. The coefficient of x^n in the expansion of $\left(\frac{1}{1-x}\right)\left(\frac{1}{3-x}\right)$ is
(a) $\frac{3^{n+1}-1}{2 \cdot 3^{n+1}}$ (b) $\frac{3^{n+1}-1}{3^{n+1}}$ (c) $2\left(\frac{3^{n+1}-1}{3^{n+1}}\right)$ (d) None
5. The co-efficient of the middle term in the expansion of $(1+x)^{2n}$ is
(a) $2^n C_n$ (b) $\frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{n!} 2^n$
(c) $2 \cdot 6 \dots (4n-2)$ (d) None of these
6. The coefficient of $x^n y^n$ in the expansion of $\{(1+x)(1+y)(x+y)\}^n$ is -
(a) $\sum_{r=0}^n C_r^2$ (b) $\sum_{r=0}^n C_r^3$ (c) $\sum_{r+s=0}^n C_r^n C_s^2$ (d) None
7. The sum of coefficients in $(x+2y+z)^{10} =$
(a) 2^{10} (b) 3^{10} (c) 4^{10} (d) None of these
8. If $\frac{T_2}{T_3}$ in the expansion of $(a+b)^n$ and $\frac{T_3}{T_4}$ in the expansion of $(a+b)^{n+3}$ are equal, then $n =$
(a) 3 (b) 4 (c) 5 (d) 6
9. The total number of terms in the expansion of $(x+a)^{100} + (x-a)^{100}$ after simplification is
(a) 50 (b) 202 (c) 51 (d) None of these
10. The value of $2({}^nC_0) + \frac{3}{2}({}^nC_1) + \frac{4}{3}({}^nC_2) + \frac{5}{4}({}^nC_3) \dots$ is -
(a) $\frac{2^n(1-n)-1}{n+1}$ (b) $\frac{2^n(n+3)-1}{n+1}$
11. Integral part of $(5\sqrt{5} + 11)^{2n+1}$ is -
(a) Even (b) Odd
(c) Even or odd depending upon the value of n (d) Cannot be determined
12. The value of ${}^{47}C_4 + \sum_{r=1}^5 ({}^{52-r}C_3)$
(a) ${}^{53}C_3$ (b) ${}^{52}C_4$ (c) ${}^{52}C_5$ (d) None of these
13. If the middle term in the expansion of $\left(\frac{1}{x} + x^{\log_2 x}\right)^5$ is 40, then x equals-
(a) $1/\sqrt{2}, 2$ (b) $\sqrt{2}, 4$ (c) $1/\sqrt{2}, 4$ (d) $\sqrt{2}, 1/\sqrt{2}$
14. If sum of the coefficients in expansion of $(1+x)^n$ is 4096 then value of largest binomial coefficients is -
(a) 792 (b) 924 (c) 462 (d) None of these
15. If $(1+x-2x^2)^6 = 1 + C_1x + C_2x^2 + \dots + C_{12}x^{12}$ then the value of $C_2 + C_4 + C_6 + \dots + C_{12}$ is
(a) 30 (b) 32 (c) 31 (d) None of these
16. If $(3\sqrt{3}+5)^7 = P + F$, where P is an integer and F is a proper fraction, then $F \cdot (P+F)$ is equal to -
(a) 3^7 (b) 2^6 (c) 3^6 (d) 2^7
17. If the coefficients of 2^{nd} , 3^{rd} and 4^{th} terms in the expansion of $(1+x)^{2n}$ are in A.P., then $2n^2 - 9n + 8 =$
(a) 0 (b) 1 (c) -1 (d) None
18. If the value of x is so small that x^2 and higher power can be neglected, then $\frac{\sqrt{1+x} + \sqrt[3]{(1-x)^2}}{1+x+\sqrt{1+x}}$ is equal to
(a) $1 + \frac{5}{6}x$ (b) $1 - \frac{5}{6}x$ (c) $1 + \frac{2}{3}x$ (d) $1 - \frac{2}{3}x$
19. Let $E = (2n+1)(2n+3)(2n+5)\dots(4n-3)(4n-1)$ where $n > 1$. then $2^n E$ is divisible by
(a) ${}^{2n}C_n$ (b) ${}^{4n}C_{2n}$ (c) ${}^{2n}C_{n/2}$ (d) ${}^{4n}C_{n/2}$
20. If $C_r = \binom{n}{r}$, then $\sum_{r=1}^n (-1)^{r-1} \frac{C_r}{r} =$

- (a) 0 (b) 1 (c) $\sum_{r=1}^n \frac{1}{r}$ (d) $\sum_{r=1}^n \frac{(-1)^{r-1}}{r}$

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